

STA322 - Project 1

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Introduction

hi

```
library(survey)
```

Loading required package: grid

Loading required package: Matrix

Loading required package: survival

Attaching package: 'survey'

The following object is masked from 'package:graphics':

dotchart

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.3      v tidyr      1.3.1
v purrr      1.0.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
x tidyr::expand() masks Matrix::expand()
x dplyr::filter() masks stats::filter()
x dplyr::lag() masks stats::lag()
x tidyr::pack() masks Matrix::pack()
x tidyr::unpack() masks Matrix::unpack()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(pps)
```

- PPS sampling without replacement was used, with each of the 41 Trinity departments treated as a primary sampling unit.
- A department's measure of size, (M_i), was its number of faculty. Thus, departments with more faculty had a larger probability of selection.
- We aimed for a sample size of 100, so we chose to do a two-stage PPS cluster sample from 20 departments and do a SRS of 5 faculty from each department selected.
- A seed of 919 was set so that the sample can be reproduced
- ***Although departments were selected without replacement, we calculated the with-replacement variance estimator. This treats the selected departments as if they had been independently sampled with replacement and provides a conservative variance approximation.

```
# load and reorganize data
```

```
trinity <- read_csv("trinity_faculty.csv")
```

```
Rows: 41 Columns: 5
```

```
-- Column specification -----
Delimiter: ","
chr (1): Dept name
dbl (4): Stratum #, # of faculty, Proportional allocation, Rounded allocation
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
trinity <- trinity |>
  select(stratum = "Stratum #",
         department = "Dept name",
         M_i = "# of faculty")
```

```

n_dept <- 20 # number of departments to select

trinity <- trinity[sample(1:nrow(trinity)), ] # randomize order of rows since systematic sampling

# systematic PPS sampling

t_x <- sum(trinity$M_i) # total number of faculty across all departments

trinity$pi_i <- n_dept * trinity$M_i / t_x # first-order inclusion probability for each department

max(trinity$pi_i) # check assumption that  $n_{xi}/t_x \leq 1$  for all  $i$ . it's not...

```

```
[1] 1.378151
```

```

# check which department p_i > 1 --> treat as certainty cases
trinity |>
  filter(pi_i > 1) |>
  select(department, M_i, pi_i)

```

```

# A tibble: 6 x 3
  department      M_i pi_i
  <chr>          <dbl> <dbl>
1 Economics      81  1.36
2 Physics        60  1.01
3 Romance Studies 66  1.11
4 Music          62  1.04
5 Biology        77  1.29
6 Mathematics    82  1.38

```

```

# dealing with certainty cases

# separate initial certainty and non-certainty departments
certainty <- trinity |>
  filter(pi_i >= 1) |>
  mutate(pi_i = 1)

remaining <- trinity |>
  filter(pi_i < 1)

n_remaining <- n_dept - nrow(certainty) # calculate number of additional departments to select

```

```

remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i) # recalculate pi_i for re
# if recalculation creates additional certainty PSUs, repeat until all remaining probabilities
while (any(remaining$pi_i >= 1)) {
  new_certainty <- remaining |>
    filter(pi_i >= 1) |>
    mutate(pi_i = 1)
  certainty <- bind_rows(certainty, new_certainty)
  remaining <- remaining |>
    filter(pi_i < 1)
  n_remaining <- n_dept - nrow(certainty)
  remaining$pi_i <- n_remaining * remaining$M_i / sum(remaining$M_i)
}

# results check
certainty |>
  select(department, M_i, pi_i) # 7 certainty cases identified (1 additional throughout reca

```

```

# A tibble: 7 x 3
  department      M_i pi_i
  <chr>          <dbl> <dbl>
1 Economics      81     1
2 Physics        60     1
3 Romance Studies 66     1
4 Music          62     1
5 Biology        77     1
6 Mathematics    82     1
7 Psychology & Neuroscience 59     1

```

```
nrow(certainty)
```

```
[1] 7
```

```
n_remaining
```

```
[1] 13
```

```
max(remaining$pi_i)
```

```
[1] 0.9246088
```

```
set.seed(919)
# select remaining cases with pps & add certainty cases

selected_depts <- ppss(remaining$M_i, n_remaining) # returns numbers of selected rows

pps_remaining <- remaining[selected_depts, ] # keep selected departments

pps_sample <- bind_rows(certainty, pps_remaining) # combine certainty psus with pps-selected

nrow(pps_sample) # check 20 departments were selected
```

```
[1] 20
```

```
# view selected departments
pps_sample |>
  select(department, M_i, pi_i)
```

```
# A tibble: 20 x 3
```

	department	M_i	pi_i
	<chr>	<dbl>	<dbl>
1	Economics	81	1
2	Physics	60	1
3	Romance Studies	66	1
4	Music	62	1
5	Biology	77	1
6	Mathematics	82	1
7	Psychology & Neuroscience	59	1
8	Political Science	43	0.795
9	Cultural Anthropology	23	0.425
10	Chemistry	36	0.666
11	African & African American Studies	19	0.351
12	Education	11	0.203
13	Art, Art History & Visual Studies	44	0.814
14	Asian & Middle Eastern Studies	45	0.832
15	Computer Science	48	0.888
16	English	50	0.925
17	Sociology	35	0.647
18	Classical Studies	19	0.351
19	History	47	0.869
20	Literature	25	0.462

```
# stage 2 srs

# srs 5 faculty from each department
pps_sample$faculty_numbers <- lapply(
  pps_sample$M_i,
  function(M_i) sample(1:M_i, 5)
)

# view numbers of faculty selected
pps_sample |>
  select(department, M_i, pi_i, faculty_numbers)
```

```
# A tibble: 20 x 4
  department          M_i pi_i faculty_numbers
  <chr>          <dbl> <dbl> <list>
1 Economics      81 1    <int [5]>
2 Physics        60 1    <int [5]>
3 Romance Studies 66 1    <int [5]>
4 Music          62 1    <int [5]>
5 Biology        77 1    <int [5]>
6 Mathematics    82 1    <int [5]>
7 Psychology & Neuroscience 59 1    <int [5]>
8 Political Science 43 0.795 <int [5]>
9 Cultural Anthropology 23 0.425 <int [5]>
10 Chemistry     36 0.666 <int [5]>
11 African & African American Studies 19 0.351 <int [5]>
12 Education     11 0.203 <int [5]>
13 Art, Art History & Visual Studies 44 0.814 <int [5]>
14 Asian & Middle Eastern Studies 45 0.832 <int [5]>
15 Computer Science 48 0.888 <int [5]>
16 English       50 0.925 <int [5]>
17 Sociology     35 0.647 <int [5]>
18 Classical Studies 19 0.351 <int [5]>
19 History       47 0.869 <int [5]>
20 Literature    25 0.462 <int [5]>
```

```
samples <- read.csv("pps_sample.csv")
```

Question 1. What is an estimate of the fraction of faculty who are women?

#|label: women-prop

Question 2. What fraction of faculty have at least one grant that is ongoing in 2026?

Question 3. Among faculty who have a Ph.D. degree, what is the average number of years since they received the Ph.D. degree?

Question 4.

a)

b) Did this faculty receive their undergraduate degree outside of the US?